

VentureMiner AI

Scoring Engine

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Table of Contents

Document 16 — Scoring Engine

Table of Contents

1. Purpose & Scope
2. Scoring principles
3. Default rubric
4. Custom rubrics
5. Scoring algorithm
6. Sub-scoring
7. Rationale
8. Confidence
9. Calibration
10. Re-scoring & history
11. Failure modes
12. Appendix
 - 12.1 Revision history
 - 12.2 Cross-references

Document 16 — Scoring Engine

The scoring engine turns evidence into a single, defensible number. This document defines the default rubric, the scoring algorithm, the rationale discipline, and the calibration methodology.

Table of Contents

1. Purpose & Scope
2. Scoring principles
3. Default rubric
4. Custom rubrics
5. Scoring algorithm
6. Sub-scoring
7. Rationale
8. Confidence
9. Calibration
10. Re-scoring & history
11. Failure modes
12. Appendix

1. Purpose & Scope

This document is the contract for the scoring engine — the system that consumes evidence and a rubric and produces a defensible score with full traceability.

2. Scoring principles

- Transparent. The rubric is editable; the formula is exposed.
- Cited. Every sub-score has at least one contributing evidence citation.
- Calibrated. Scores are tested against expert judgments monthly.
- Versioned. A score is always bound to a specific rubric version.
- Composable. Sub-scores can be combined in user-defined ways.

3. Default rubric

Market size	20%	TAM/SAM/SOM band
Growth	15%	YoY/CAGR
Demand	20%	Search/social/intent
Buildability	15%	Time + cost to MVP
Defensibility	15%	Differentiation, moat
AI-fit	15%	Natural LLM advantage

4. Custom rubrics

- Workspace admins can edit:
- Add / remove / rename dimensions.
- Change weights (must sum to 100).
- Add sub-criteria to a dimension.
- Override description text.
- Every change is a new rubric version.
- Scores from old versions remain visible.

5. Scoring algorithm

For each dimension:

1. Retrieve relevant evidence (RAG, scoped to opportunity).
2. LLM judge produces a sub-score (0–10) + rationale + confidence.
3. Verifier audits the sub-score: - Rationale references real evidence. - Sub-score is consistent with rationale. - No policy violation.

Total score = $\sum (\text{weight}_i \times \text{sub_score}_i)$, normalized to 0–10.

6. Sub-scoring

A dimension can have sub-criteria. The dimension score is:

$$\text{dim_score} = \sum (\text{sub_weight}_j \times \text{sub_score}_j)$$

The total score is unchanged (sum of dim_weights × dim_scores).

7. Rationale

Every sub-score ships with:

- 1–3 sentences explaining the score.
- Citations to the evidence that drove the score.
- A confidence grade (low / med / high).

Rationales are user-visible; they are the trust surface.

8. Confidence

Confidence is graded by:

- Evidence count — more sources → higher confidence.
- Evidence quality — primary sources rank higher than secondary.
- Recency — recent data boosts confidence.
- Convergence — multiple sources agreeing boost confidence.

- Calibration set: 200 scored opportunities reviewed by 3 experts.
- Monthly: the engine scores the set; mean absolute error (MAE) is measured.
- Threshold: MAE > 1.0 triggers a rubric review.
- Drift detection: weekly tracking of user feedback (acceptance, edits); significant drift triggers recalibration.

10. Re-scoring & history

- Re-score on new evidence is automatic (subscribed to validation.completed).
- History is preserved; users can view the score timeline.
- Diff is shown when the score changes by > 0.5.

11. Failure modes

LLM judge unavailable	Retry; fallback model.
Evidence missing for a dimension	Sub-score = null; dimension excluded from total; surface.
Rubric invalid (weights ≠ 100)	Reject; admin must fix.
Verifier rejects twice	Mark dimension unverified; surface.
Calibration drift	Trigger rubric review.

12. Appendix

12.1 Revision history

v0.5	2026-07-20	Doc Team	All sections drafted
v1.0	2026-07-20	Doc Team	First approved version

12.2 Cross-references

- Agent Specs: Document 09 §14.
- Research Pipeline: Document 15.
- Report Generation: Document 17.

End of Document 16 — Scoring Engine. A score without rationale is not a score — it is a guess.